

**RADEON HACKATHON 2026 · TRACK 2: AGENTIC AI**

# Screen-AI

A fully local, offline AI operator for your PC — controlled from your phone.

**100% LOCAL / OFFLINE**

**AMD ROCm ACCELERATED**

**RUNS ON 4GB RAM**

Phone = command · approval · unlock

PC = perception · planning · action

## THE PROBLEM

# Today's AI agents don't respect your machine — or your privacy

- ▶ Cloud agents stream your screen and keystrokes to remote servers
- ▶ Always-on internet, accounts, and subscriptions required
- ▶ Network latency on every click — no offline fallback
- ▶ Heavy models don't fit on ordinary 4–8GB laptops
- ▶ Risky actions execute without your explicit consent

## THE SOLUTION

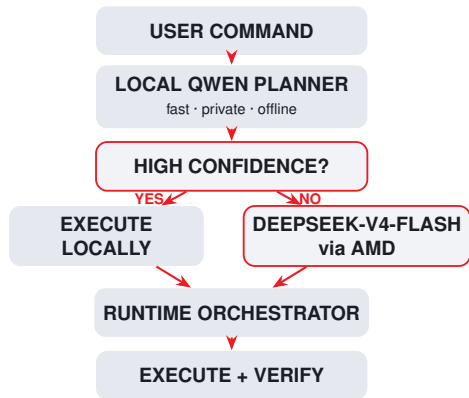
# An agentic operator that lives entirely on your PC

- ▶ **Sees the screen:** Windows UIA + OpenCV detect buttons, inputs, tabs, menus
- ▶ **Acts locally:** click by text, fill forms, automate the browser, manage files
- ▶ **Phone is the remote:** send commands, approve risky actions, emergency stop
- ▶ **Hybrid AI brain:** local Qwen 1.5B GGUF plans first — hard intents escalate to DeepSeek-Flash via AMD

## WHY IT WINS

- ▶ Privacy — nothing leaves the machine by default
- ▶ Offline — zero cloud dependency for local tasks
- ▶ Lightweight — RAM-aware tiers fit 4GB laptops
- ▶ Safe — approval, quarantine, kill switch

# Hybrid cognitive engine — local first, cloud when it counts



## WHY AN EXTERNAL API AT ALL?

- ▶ **Local first:** the on-device Qwen planner handles most commands instantly — offline, private, zero latency
- ▶ **Escalate only when unsure:** ambiguous or multi-step intents route to DeepSeek-Flash reasoning API hosted on AMD cloud
- ▶ **Why not a bigger local model?** it breaks the 4GB RAM budget — a hybrid router beats a giant model on a laptop
- ▶ **Same safety net:** cloud-enhanced plans still pass through the Runtime Orchestrator and phone approval

## CAPABILITIES

# Everything an operator needs — built in

01

### Mobile Remote Control

Text commands from phone to PC over REST / WebSocket

02

### Screen Perception

UIA + OpenCV detect buttons, inputs, tabs, menus

03

### Screen Control

Click elements by text, scan, verify every action

04

### Browser Automation

Search, open sites, fill forms, download files (Playwright)

05

### Password Vault

AES-256-GCM + Argon2id encrypted credential storage

06

### Phone Approval

Risky actions need one-tap approval before execution

07

### File Quarantine

Deletes are reversible — never destructive

08

### Emergency Stop

One tap halts all operations instantly

## HOW IT WORKS

# From your phone to the desktop — in milliseconds



**01**

### Security Layer

Risk scoring, permissions, vault, approvals

**02**

### Tier Manager

RAM-aware model tier selection

**03**

### Resource Budget

psutil live memory measurement

**04**

### Screen Scanner

UIA + OpenCV perception

## Three modules, one operator

### Screen Scanner

`screen_element_scanner/`

The eyes — UIA + OpenCV  
screen perception of the Windows UI

### Distillation Pipeline

`hackathon_ui_operator_distill/`

Cloud teacher → local YOLOv8n  
INT8 ONNX student model

### AI PC Operator

`ai_pc_operator/`

FastAPI server + mobile remote  
+ agent brain + runtime engine

# RAM-aware tiered loading — built for 4GB laptops

## **TIER 0**

### **Resident**

UIA, OpenCV, rule planner — always in RAM

## **TIER 1**

### **Warm**

OCR + YOLO detector — lazy-loaded when RAM allows

## **TIER 2**

### **SSD Cold**

Heavy models — on disk, loaded on demand

## **TIER 3**

### **SSD Off**

Too heavy for the current RAM profile

## **LOW-MEMORY PLAYBOOK**

Memory-mapped GGUF files · 512-token context · browser idle-evicts after 120s · Qwen LLM stays on SSD by default · OCR/detector load only if RAM allows



**SECURITY FIRST**

## Local doesn't mean careless

### **AES-256-GCM Vault**

Authenticated encryption for all stored credentials

### **Argon2id**

Memory-hard key derivation against brute force

### **X25519 QR Pairing**

Phone–PC key exchange via QR — no passwords typed

### **HMAC Signatures**

Every request authenticated and tamper-proof

### **Phone Approval**

Risky actions blocked until approved on phone

### **Quarantine + E-Stop**

Reversible deletes and a one-tap halt for everything

# Heavy teacher in the cloud, tiny student on the laptop

## CLOUD — TEACHER PHASE

- ▶ Collect UI screenshots
- ▶ OmniParser v2 generates teacher labels
- ▶ Build YOLO training dataset
- ▶ Distill into YOLOv8n → INT8 ONNX student

## LAPTOP — LIGHT RUNTIME

- ▶ Windows UI Automation + OpenCV
- ▶ PaddleOCR for on-screen text
- ▶ Tiny INT8 ONNX detector (YOLOv8n)
- ▶ Local Qwen 1.5B GGUF planner via llama.cpp

UNDER THE HOOD

## Tech stack

### BACKEND

Python 3.10+ · FastAPI · aiosqlite · psutil · Playwright · cryptography · pyautogui

### FRONTEND

Vanilla JS / CSS / HTML · WebSocket · QRCode.js

### ML / TRAINING

Ultralytics YOLOv8n · ONNX Runtime · llama.cpp (GGUF) · PaddleOCR

### SECURITY

AES-256-GCM · Argon2id · X25519 (QR pairing) · HMAC signatures

LIVE DEMO

# Up and running in under a minute

## RUN IT

```
.\ai_pc_operator\start.bat  
python backend/scripts/start_https.py  
python backend/scripts/download_models.py
```

## OPEN IT

- ▶ localhost:8000/remote/pair.html — PC shows QR
- ▶ localhost:8000/remote/index.html — phone remote
- ▶ localhost:8443 — HTTPS for camera QR scanning

1 · Start the server

2 · PC displays pairing QR

3 · Phone scans & pairs (X25519)

4 · Command, approve, done

# Screen-AI

Local. Private. Agentic.

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Thank you — questions welcome